

Recent Developments and Future Directions of Wearable Skin Biosignal Sensors

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This review article explores the transformative advancements in wearable biosignal sensors powered by machine learning, focusing on four notable biosignals: electrocardiogram (ECG), electromyogram (EMG), electroencephalogram (EEG), and photoplethysmogram (PPG). The integration of machine learning with these biosignals has led to remarkable breakthroughs in various medical monitoring and human–machine interface applications. For ECG, machine learning enables automated heartbeat classification and accurate disease detection, improving cardiac healthcare with early diagnosis and personalized interventions. EMG technology, combined with machine learning, facilitates real-time prediction and classification of human motions, revolutionizing applications in sports medicine, rehabilitation, prosthetics, and virtual reality interfaces. EEG analysis powered by machine learning goes beyond traditional clinical applications, enabling brain activity understanding in psychology, neurology, and human–computer interaction, and holds promise in brain–computer interfaces. PPG, augmented with machine learning, has shown exceptional progress in diagnosing and monitoring cardiovascular and respiratory disorders, offering non-invasive and accurate healthcare solutions. These integrated technologies, powered by machine learning, open new avenues for medical monitoring and human–machine interaction, shaping the future of healthcare.

1. Introduction

Biomedical devices are highly specialized instruments that have been developed to diagnose, monitor, and treat a wide range

of medical conditions. They are an integral part of modern medicine, providing healthcare professionals and researchers with valuable insights that can help enhance patient outcomes. The field of biomedical devices is inherently interdisciplinary, requiring collaboration across multiple disciplines, including biology, medicine, engineering, physics, and materials science.^[1] This range of devices encompasses a vast array of instruments, from basic sensory devices such as thermometers and blood glucose meters to advanced surgical and implantable devices like surgical robots and prosthetics. Advancements in imaging technologies have led to the development of devices such as X-ray, magnetic resonance imaging (MRI), and computed tomography scanners, which can visualize internal structures and diagnose diseases.^[2] Similarly, advances in mechanical engineering have led to the development of devices that can replace or support body organs, such as mechanical hearts, artificial joints, and prosthetics.^[3] With the rapid growth of biology and materials science, cutting-edge

biomedical devices are extensively studied. For instance, 3D-printed artificial organs, microbots, and neural probes are emerging as new types of devices with significant potential to improve patient outcomes.^[4] These devices hold great promise for the future of healthcare, and ongoing research and development in this field will continue to drive innovation and enhance patient care.

In recent years, the field of electronic engineering has witnessed remarkable advancements. For instance, in 2001, microprocessors were produced using a 130 nm process, while today's processors are manufactured using only a 3 nm process.^[5] The increasing demand for wireless devices has also spurred advances in wireless technology. Furthermore, recent advancements in biomedical devices have led to the creation of wearable devices like fitness trackers and smartwatches, which can monitor vital signs and provide real-time feedback on health and wellness.^[6] In summary, the development and use of biomedical devices have transformed modern healthcare, providing clinicians and researchers with powerful tools to diagnose, monitor, and treat medical conditions. The ongoing innovation and advancement of biomedical devices hold great promise for improving patient outcomes and advancing medical knowledge in the future.

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 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adsr.202300118>

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DOI: 10.1002/adsr.202300118

Machine learning is an artificial intelligence field that develops algorithms to enable computers to learn from data and make predictions or decisions without explicit programming. It can analyze large datasets, identify patterns, and improve performance through feedback. It has broad applications, including image/speech recognition, recommendation systems, and autonomous vehicles. It is a rapidly growing field that has transformative potential. Machine learning has had a significant impact on the development of biomedical healthcare devices. It has enabled the creation of more accurate and efficient diagnostic tools, such as image analysis systems that can identify tumors in medical images with high precision.^[7,8] Machine learning has also been used to develop predictive models for patient outcomes and to improve the accuracy of medical sensors, such as those used for monitoring vital signs.^[9] Additionally, machine learning has facilitated the development of personalized medicine by enabling the analysis of large datasets to identify individual patient needs and create tailored treatment plans. Overall, machine learning has contributed to the development of more effective and efficient biomedical healthcare devices, which can ultimately improve patient outcomes and save lives.

Biosignals are the electrical signals generated by the physiological activities of living organisms, such as the heart, muscles, and brain.^[10] Biosignals can provide valuable information about the subjects' health status, emotions, and cognitive functions. However, measuring and analyzing biosignals is challenging due to their complexity, variability, and noise.^[11] In this paper, we focused on four types of biosignals: electrocardiogram (ECG), electromyogram (EMG), electroencephalogram (EEG), and photoplethysmogram (PPG). We discuss their features, applications, and challenges in the context of wearable biosensors and Internet of Things (IoT) systems.^[12] We also reviewed the recent advances in machine learning techniques that can assist the wearable biosensors in processing and interpreting the biosignals. We aim to provide a comprehensive overview of the current state-of-the-art and future directions of biosignal-based wearable biosensors and IoT systems (Figure 1).

2. Biosignals and Their Features

2.1. Electrocardiogram (ECG)

ECG is a technology for measuring cardiac voltage signals over time. ECG is a non-invasive procedure that could diagnose various heart conditions, including cardiovascular malformations such as arrhythmia.^[13] ECG plot involves the waveform segments; P wave, QRS complex, and T wave (Figure 2a). Each of these segments provides important information about the health and function of the heart. For instance, the P wave reflects the specific sequence of atrial activation, while the QRS complex represents ventricular depolarization, and the T wave reflects ventricular repolarization.^[14,15] By measuring the duration and amplitude of these segments, healthcare professionals can assess the heart's overall health and diagnose various heart conditions such as arrhythmias and conduction abnormalities. ECG patterns could be characterized by both temporal features and spectral features. Temporal features of ECG signals are related to the timing and duration of various components of ECG signals over the time domain. The most fundamental temporal feature

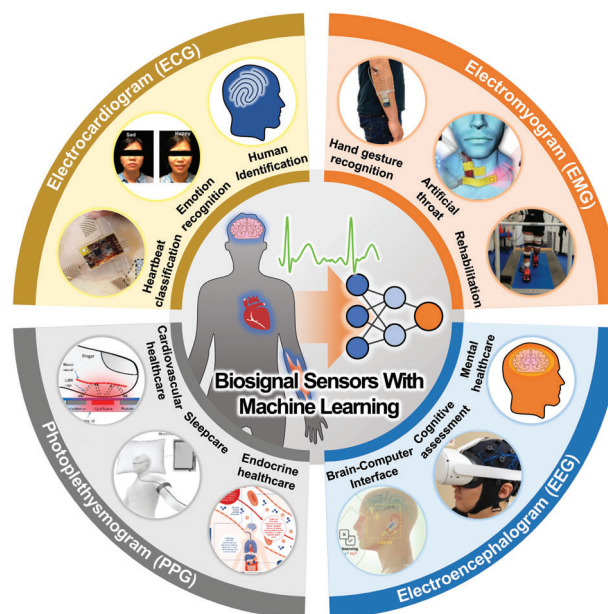


Figure 1. Overall schematic of wearable biosignal sensors incorporated with machine learning techniques. ECG analysis for heartbeat classification. Reproduced under the terms of the CC-BY license.^[78] Copyright 2019, the authors, published by John Wiley and Sons. ECG analysis for Emotion recognition. Reproduced with permission.^[83] Copyright 2020, American Chemical Society. EMG analysis for hand gesture recognition. Reproduced with permission.^[146] Copyright 2020, Springer Nature. EMG analysis for the artificial throat. Reproduced with permission.^[94] Copyright 2022, Elsevier. EMG analysis for rehabilitation. Reproduced under the terms of the CC-BY license.^[154] Copyright 2022, the authors, published by MDPI. EEG analysis for brain-computer interface. Reproduced under the terms of the CC-BY license.^[97] Copyright 2022, the authors, published by Springer Nature. EEG analysis for cognitive assessment. Reproduced with permission.^[177] Copyright 2022, Elsevier. PPG analysis for cardiovascular healthcare. Reproduced under the terms of the CC-BY license.^[206] Copyright 2019, the authors, published by MDPI. PPG analysis for sleep care. Reproduced with permission.^[207] Copyright 2023, Elsevier. PPG analysis for endocrine healthcare. Reproduced with permission.^[208] Copyright 2020, Springer Nature.

is heart rate, which is often calculated by the RR interval between consecutive R peaks. Heart rate variability (HRV), reflecting variations in these RR intervals over time, emerges as a valuable temporal feature in ECG analysis.^[16] HRV serves as an indicator of cardiac health and provides insights into the activity of the autonomic nervous system.^[17] Furthermore, PR interval (or PQ interval), QRS duration, QT interval, and ST segment could be utilized as important temporal features for a comprehensive assessment of cardiac function and pathology (Figure 2a).^[18–21] In addition to the temporal features, spectral features also provide information about the frequency content of the ECG signals. By employing techniques such as Fourier analysis and wavelet transforms, ECG signals can be transformed into their frequency domain representations, revealing the distribution of power across these spectral components. In ECG analysis, spectral features refer to distinct frequency bands, including low-frequency (LF; 0.04 to 0.15 Hz), high frequency (HF; 0.15 to 4 Hz), and very low frequency (VLF; 0.0033 to 0.04 Hz), and ultralow frequency

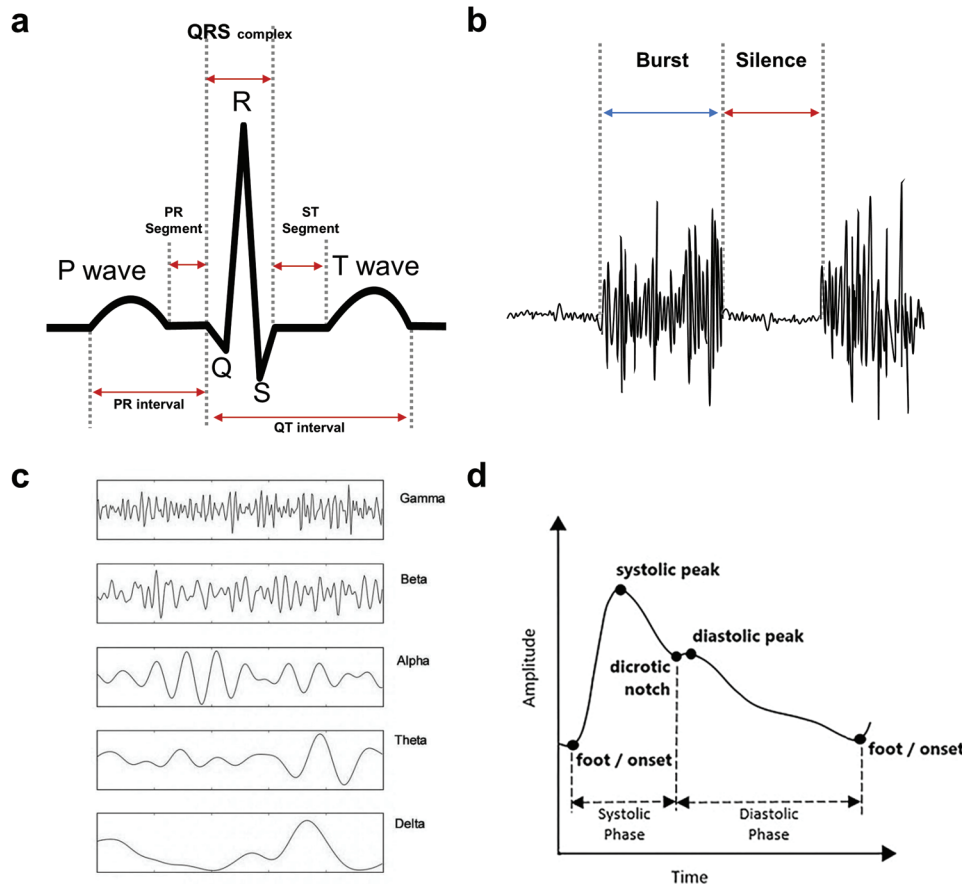


Figure 2. Biosignals and their features. a) ECG waveform segments and temporal features. b) EMG burst period and silence period. c) EEG signals in different frequency bands, delta (0.1 to 4 Hz), theta (4 to 8 Hz), alpha (8 to 13 Hz), beta (13 to 30 Hz), and gamma (30 to 100 Hz) waves. Reproduced with permission.^[209] Copyright 2019, Elsevier. d) PPG waveform with distinct phases. Reproduced under the terms of the CC-BY license.^[45] Copyright 2022, the authors, published by Frontiers Media S.A.

(ULF; 0.0033–0.04 Hz) components, which encode essential information about cardiac rhythms and autonomic nervous system activity.^[22] By isolating and characterizing these dominant frequencies, spectral analysis unveils the intricate interplay of physiological processes governing the heart's behavior. For example, Otsuka et al. conducted a 6-month study to examine how the cardiovascular system of astronauts adapts to the microgravity environment of the International Space Station.^[22] They analyzed changes in the power-law scaling β of HRV within different frequency bands, including HF, LF, VLF, and ULF components of ECG signals.

2.2. Electromyogram (EMG)

EMG is a technology used to measure the electrical signals produced by muscles and nerves. It can detect the electrical potential generated by activated muscle cells. EMG has been widely applied in various fields such as sports medicine,^[23,24] rehabilitation,^[25,26] prosthetics,^[27] and ergonomics.^[28] The technology is also used in the diagnosis of neuromuscular disorders, such as muscular dystrophy,^[29] amyotrophic lateral sclerosis,^[30]

and fasciculation.^[31] The analysis of EMG signals requires a multi-step process to decipher the intricacies of muscle electrical activity. It initiates with the acquisition of raw EMG data, capturing both the dynamic bursts of muscle electrical activity and the underlying baseline periods of silence (Figure 2b).^[32] The baseline voltage represents the resting state of the muscle, serving as a reference point to distinguish periods of muscle activity from periods of quiescence. burst activity in EMG represents a clustered firing pattern of motor units, with multiple motor unit action potentials occurring closely together in time.^[33] To enhance signal quality and relevance, preprocessing steps entail filtering, which removes noise while emphasizing specific frequency bands, and rectification, which transforms the signal into a unipolar form, often through half-wave or full-wave rectification.^[34] Envelope extraction is then employed, producing smoothed representations that encapsulate the signal's amplitude variations within distinct frequency bands.^[35] This enveloped data serves as the foundation for subsequent analyses. Root mean square analysis quantifies overall signal magnitude and amplitude fluctuations within specified windows, facilitating assessments of muscle strength, endurance, and coordination.

2.3. Electroencephalogram (EEG)

EEG is a non-invasive technique used to measure and record the brain's electrical activity. Traditionally, EEG has been employed in clinical settings for diagnosing and monitoring neurological disorders, such as epilepsy and sleep disorders.

EEG signal analysis is a fundamental process integral to extracting valuable insights from recordings of brain electrical activity. The process involves several essential steps, commencing with data acquisition through electrodes placed on the scalp to capture electrical signals originating from brain neurons. In the preprocessing stage, filters are applied to eliminate noise and emphasize frequency bands of interest, such as delta (0.1 to 4 Hz), theta (4 to 8 Hz), alpha (8 to 13 Hz), beta (13 to 30 Hz), and gamma (30 to 100 Hz) waves (Figure 2c).^[36] Delta waves typically have a high amplitude of 20 to 200 microvolts (μV), theta waves are generally $>20 \mu\text{V}$, alpha waves often range from 30 to 50 μV , beta waves typically range from 5 to 30 μV , and gamma waves usually have amplitudes $<5 \mu\text{V}$.^[37] Additionally, artifacts caused by eye blinks, muscle activity, or other sources are detected and removed by the statistical algorithm such as regression method, principal component analysis, and independent component analysis.^[38] Spectral analysis constitutes another vital facet of EEG analysis, involving the comprehensive examination of power spectral density to assess the distribution of power across distinct frequency bands, thereby unveiling intricate brain activity patterns.^[39] Additionally, event-related potentials play a pivotal role by identifying and analyzing brain responses specifically tied to particular stimuli or events.^[40] This collective process not only sheds light on the intricacies of neural activity but also contributes significantly to our understanding of cognitive processes and brain function.

2.4. Photoplethysmogram (PPG)

A plethysmogram is a graphical representation of changes in the dimensions or capacity of an organ or bodily region over a period. It is commonly used to quantify alterations in blood flow or volume in various body regions. Plethysmography is a non-invasive technique that can provide critical information regarding an individual's circulatory system, serving as a diagnostic and monitoring tool for cardiovascular or respiratory disorders. Typically, plethysmography employs an inflatable cuff to detect changes in fluid flow. In contrast, PPG uses light to measure changes in blood volume in a specific area. A light-emitting diode (LED) emits light through the skin, and a photodetector (PD) detects the light penetrating through the tissue. Therefore, PPG has gained widespread use in medical settings due to its convenience and increased accuracy.^[41–43]

The PPG waveform comprises distinct phases: the systolic phase, representing the contraction of the left ventricle and the ejection of blood into the arteries; the diastolic phase, indicating the relaxation of the heart and the refilling of blood in the arteries; and the dicrotic notch, signifying the closure of the aortic valve (Figure 2d).^[44,45] Additionally, the PPG signal consists of a direct current steady part, reflecting baseline blood volume, and an alternating current (AC) pulsatile part, representing variations in blood volume with each heartbeat.^[46] By analyzing these compo-

nents, valuable information can be extracted, including heart rate from pulse rate in the AC component or blood pressure trends by evaluating the amplitude and morphology of the systolic and diastolic phases.^[47]

3. Wearable Biosensors and IoT System

In the pursuit of real-time monitoring of personal healthcare and clinical information based on biosignals, the seamless collection, transmission, and processing of data are paramount. Such a system must operate effortlessly, adapting to the user's activities without interruption. In contemporary healthcare, wearable biosensor designs, in conjunction with IoT systems, have emerged as a dynamic area of research, promising real-time applications that are further empowered by machine learning techniques for advanced healthcare and clinical scenarios.

This chapter delves into the evolution of device designs, ranging from traditional hospital setups to innovative flexible and stretchable patch-type devices. Additionally, we explore the intricate components that constitute IoT systems, facilitating continuous data transmission and enabling modifications, all of which contribute to a comprehensive understanding of the prevailing trends and requirements driving the realm of real-time biosensor applications.

3.1. Device Design for Biosensors

3.1.1. Hospital Setup

In a clinical hospital setting, biosensors play a pivotal role in gathering critical data for diagnoses and treatment planning. The precision, accuracy, and reliability of data collection are paramount, ensuring healthcare professionals can provide optimal care to patients.

To accurately measure the ECG signals, conventional medical equipment uses 12 or 15 leads of electrodes which are attached to the patient's chest, arms, and legs.^[48] Disposable ECG electrode pads are usually composed of Ag/AgCl electrodes, conductive gels, and adhesives. Due to its conformal contact with the skin and low impedance of the conductive gels, it ensures the accurate measurements of signals.

In EMG analysis, needle electrodes are often inserted into the muscle being tested for diagnostic purposes.^[49] For the alternative non-invasive method, surface EMG (sEMG) utilizing patch-like electrodes could be conducted.^[50] Commercial sEMG electrodes share similarities with disposable ECG electrodes, primarily using Ag/AgCl conductors, electrolyte gels, and adhesives.^[51]

In EEG, adherence to the standardized 10–20 system ensures precise placement, capturing electrical activity from specific brain regions.^[52] Conventionally, disc-shaped disposable electrodes with conductive gel have been utilized, while electrode caps or nets also could be employed to enhance comfort and reduce preparation time.^[53]

Pulse oximeters, perhaps the most well-known PPG devices used in clinical settings, are often clipped onto a patient's fingertip, earlobe, or other peripheral areas to measure oxygen saturation and pulse rate.^[54] These devices use LEDs to emit two

different wavelengths of light, typically red and infrared (IR), and detect the light's absorption through the patient's tissue.^[55] These LEDs are positioned on one side of the device, and they emit light through the patient's tissue to the PD on the other side.

For clinical purposes, all these hospital setups, including ECG, EMG, EEG, and PPG, involve external wires and machines for data processing, recording, and monitoring. Proper electrode placement in ECG, EMG, and EEG, as well as the correct device positioning on the patient's body for PPG, are crucial for accurate measurements and are typically guided by trained healthcare professionals or technicians. While hospital setups offer advantages in terms of accuracy and reliability, they are often constrained by limited portability and may not be conducive to continuous real-life monitoring with optimal comfort.

3.1.2. Portable Devices

The advent of portable biosensors has ushered in a new era of convenience and versatility in monitoring vital physiological parameters such as ECG, EMG, EEG, and PPG.

In the realm of ECG, portable single-lead ECG devices featuring thumb electrodes were developed for quick and easy heart monitoring.^[56] Nonetheless, these devices are unsuitable for continuous measurement in daily life due to the restriction of hand usage during measurement as it keeps requiring contact from both thumbs. To overcome this limitation, wearable ECG electrodes have been developed, enabling continuous data collection when combined with IoT systems. Several studies have employed armband-type electrode designs integrated with an IoT system for mobile ECG monitoring.^[57–59] Additionally, textile electrodes within garments such as T-shirts or brassieres have successfully been utilized in wearable ECG devices.^[60–62]

In the domain of EMG, Various wearable EMG sensor designs have been developed for different muscle groups and activities to enhance wearability and portability. For the analysis of upper limb muscles, armband-type devices have been reported.^[63–67] These devices can be easily worn around the arm, allowing for convenient monitoring of upper limb muscles' EMG signals. Options such as sock-type^[68] or jogging leggings-type^[69] devices have been designed for lower limb muscles. Additionally, glasses-type devices incorporating EMG electrodes attached to the temple part allow the measurement and collection of EMG signals near the temporalis muscle.^[70,71]

Portable EEG devices offer flexibility in brainwave monitoring. Wireless EEG caps with Bluetooth or Wi-Fi modules allow for untethered and comfortable brain activity tracking, making them valuable tools in real-life brain monitoring applications.^[72] Many commercial portable EEG devices show different electrode types (dry or with the conductive gel) or device shapes (head cap or headband) for diverse user experience.^[73] In addition, the ear-EEG design, placed on or in the ear, has been offered a comfortable and unobtrusive option for tracking brain activity during various activities and tasks.^[74–77]

In the domain of PPG, smartwatches equipped with PPG sensors have become ubiquitous, enabling continuous heart rate and blood oxygen level monitoring. These wearables are invaluable for fitness tracking and general health monitoring. Armband PPG monitors offer similar capabilities, making them suitable

for athletes and individuals interested in health and fitness tracking. These portable biosensors bridge the gap between clinical precision and real-world applicability, empowering individuals to take control of their health, researchers to gather data in natural settings, and healthcare providers to offer more personalized care.

3.1.3. Flexible or Stretchable Patch-Type Devices

Innovations in biosensor technology have opened a new era of flexible or stretchable patch-type devices, revolutionizing the landscape of ECG, EMG, EEG, and PPG monitoring. These devices offer conformal contact with the skin, enhancing the signal-to-noise ratio (SNR) and significantly improving wearability and user comfort. ECG, EMG, and EEG, which require the use of electrodes to detect electrical signals, share common design features for flexible/stretchable electrodes. The softness of these patch-type electrodes allows for their application on various curved surfaces of the body, leading to their successful integration in areas such as the chest for ECG,^[78–90] target muscles like the forearm for EMG,^[79–82,84,85,87–89,91–95] and scalp or ear for EEG.^[81,82,87,89,96,97]

One common approach to fabricating flexible or stretchable electrodes is employing serpentine electrodes patterned on a flexible/stretchable substrate.^[78–81,91–93,96–97] For instance, Kim et al. developed a three-channel stretchable ECG sensor that incorporated Cu serpentine electrodes with a flexible circuit board encapsulated in a soft elastomer (Ecoflex) (Figure 3a).^[78] This system ensured the conformal contact between the skin and electrodes, enabling low skin-electrode contact impedance and high SNR. Tian et al. introduced a Cr/Au serpentine electrode on a PET substrate patterned by a photolithography process.^[81] Based on a large-area fabrication, they utilized 8-channel EMG, EEG arrays with 68 electrodes, and ECG electrodes.

Another method is using conductive ink such as metal nanoparticles, nanoflakes, and nanowires, or liquid metals with stretchable substrate.^[84–88,95] For example, Reis Carneiro et al. fabricated a thin-film biosticker by directly writing the conductive ink that is a mixture of Ag flakes, eutectic gallium–indium liquid metal, and SIS hyperelastic binder (Figure 3b).^[87] It showed a lower impedance than the conventional Ag/AgCl electrode, successfully utilized as an ECG, EMG, and EEG reading electrode and circuit.

Utilizing conductive polymer as an electrode is also developed for the flexible/stretchable electrodes.^[82,83,89,90] Zhang et al. fabricated an intrinsically stretchable, adhesive, biocompatible, and conductive film by blending poly(ethylenedioxythiophene):poly(styrenesulfonate) as a conductive component, Waterborne polyurethane as an elastomeric component, and D-sorbitol as an adhesive component.^[82] They utilized this electrode to acquire ECG, EMG, and EEG signals with lower electrode-skin impedance to the Ag/AgCl gel electrode case. Another innovative approach was proposed by Yang et al., who developed a conformal and adhesive electrode based on polypyrrole and silk fibroin (Figure 3c).^[83] This electrode exhibited conformal properties and strong adhesion, even under sweaty conditions, allowing stable continuous ECG monitoring during two hours of running.

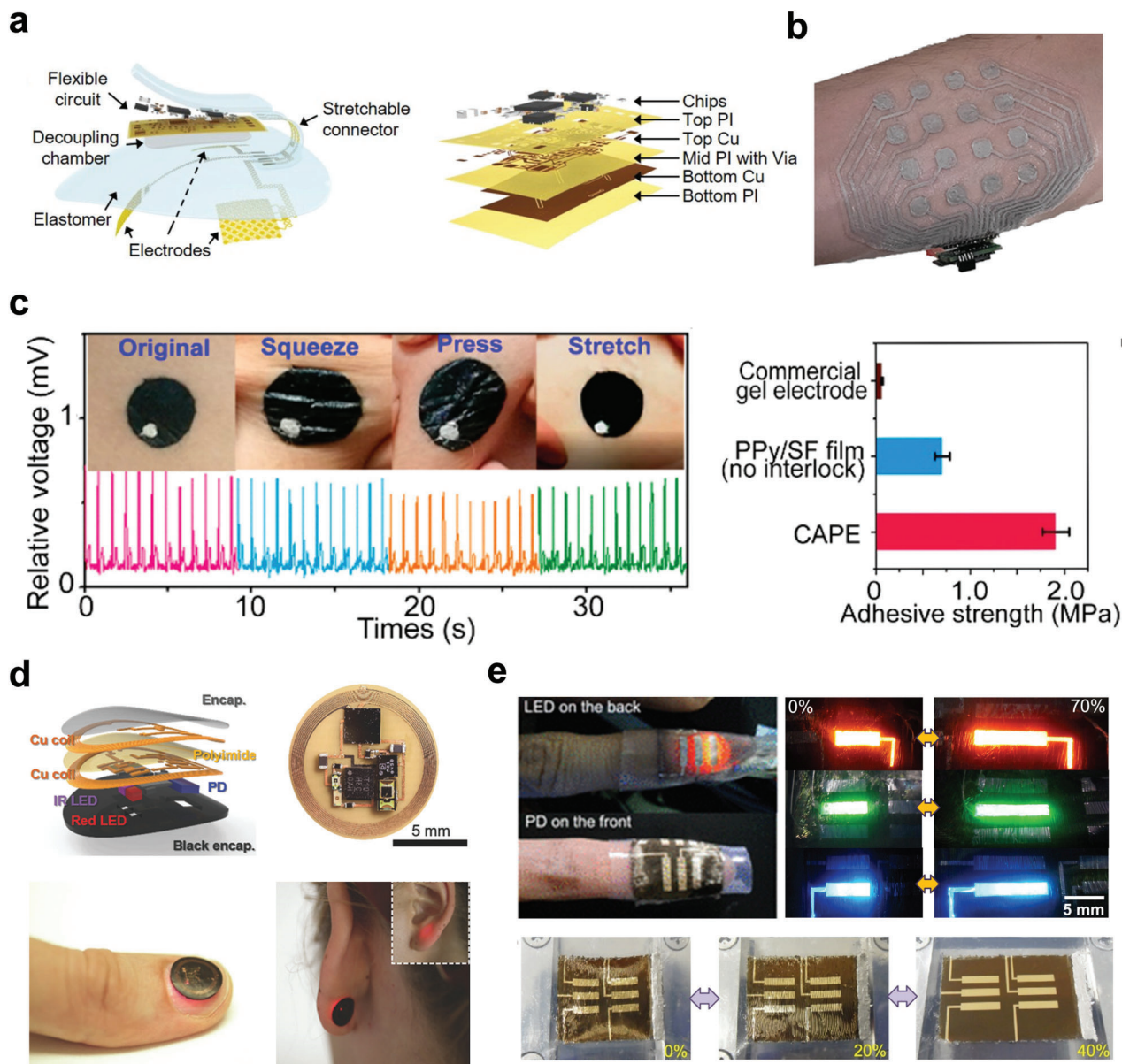


Figure 3. Diverse flexible and stretchable patch-type wearable biosensor devices. a) A three-channel stretchable ECG sensor that incorporated Cu serpentine electrodes with a flexible circuit board encapsulated in a soft elastomer. Reproduced under the terms of the CC-BY license.^[78] Copyright 2019, the authors, published by John Wiley and Sons. b) EMG patch with a 16-electrode array based on conductive ink (mixture of Ag flakes, eutectic gallium-indium, and SIS). Reproduced with permission.^[87] Copyright 2022, John Wiley and Sons. c) Stretchable ECG based on polypyrrole and silk fibroin with conformality and strong adhesion. Reproduced with permission.^[83] Copyright 2020, American Chemical Society. d) Miniaturized battery-free PPG device with red and IR LEDs, PD, near-field communication interface, loop antenna, microcontroller, and amplifiers attached to the fingernail or the back or earlobe. Reproduced with permission.^[100] Copyright 2016, John Wiley and Sons. e) Patch-type PPG sensor device with stretchable quantum dot LEDs and PDs. Reproduced with permission.^[107] Copyright 2017, American Chemical Society.

On the other hand, alternative strategies have been explored to advance patch-type PPG devices. While PPG sensors embedded within smartwatches or wristbands have reached a certain level of maturity, they are susceptible to limitations due to motion artifacts, especially during dynamic hand movements.^[98] To address this challenge, efforts have been directed toward the development of flexible and stretchable patch-type PPG devices.

One approach involves using small LED and PD components with flexible or stretchable electrode circuits.^[99–102] For example,

Kim et al. introduced a flexible miniaturized wireless battery-free PPG sensor device incorporated with red and IR LEDs, PD, near-field communication interface, loop antenna, microcontroller, and amplifiers (Figure 3d).^[100] This device successfully attached to the fingernail or the back or earlobe, wirelessly transmitted PPG signals.

Another promising approach is utilizing inherently flexible or stretchable optoelectronic components.^[98,103–107] For example, Lee et al. fabricated an organic PPG sensor with 17×7 OLED

arrays for a display capable of enduring strains up to 30%.^[103] These stretchable optoelectronic components were created by incorporating stress relief layers and utilizing stretchable micro-cracked Au interconnects to ensure their seamless attachment to the skin. Kim et al. fabricated a stretchable LED and PD that could withstand 70% and 40% external strain by the transfer printing of ultrathin quantum dot (QD) LEDs and QD PDs on a pre-strained elastomer matrix (Figure 3e).^[107] Incorporating these optoelectronic components, a patch-type PPG sensor attached to the forefinger tip was successfully operated showing a clear signal during the wrist movements.

3.2. IoT Systems for Continuous Data Transmission and Processing

The fusion of biosensors with IoT systems has ushered in a transformative era of interconnected healthcare data. These biosensors no longer function in isolation but are now integral components of a vast network, facilitating real-time data transmission and processing.

The data journey begins with the acquisition of raw sensor data, followed by dedicated hardware-driven processes for denoising, filtering, and signal amplification, all aimed at enhancing signal quality. Subsequently, the data is seamlessly transmitted through interconnection technologies such as flexible flat cables or wireless communication modules like Bluetooth and Wi-Fi. Upon reaching cloud-based servers, the data undergoes further preprocessing and in-depth analysis.

These servers are equipped with computational capabilities, enabling real-time monitoring and even machine learning algorithms for predictive analytics. This convergence of biosensors and IoT is poised to revolutionize healthcare by providing timely and actionable insights, enhancing patient care, and contributing to the development of personalized healthcare solutions.

4. Machine Learning-Assisted Wearable Biosensors

4.1. ECG with Machine Learning Techniques

Recently, the integration of machine learning techniques in ECG analysis has enabled the automated classification of ECG signals into various classes, including normal and arrhythmia. Furthermore, the incorporation of wearable ECG technology holds immense potential for enabling continuous real-time classification and diagnosis of arrhythmias and other heart diseases, leading to earlier detection and treatment interventions that can greatly benefit patients.

For the classification task, machine learning algorithms such as Support vector machines (SVM),^[108–110] k-nearest neighbors (kNN),^[111,112] and a random forest have been utilized.^[113,114] Before applying the classifier, feature extraction was employed for more accurate learning than just classifying raw ECG signals. For instance, Saini et al. extracted statistical features from raw ECG signals and sixth-level wavelet transformed ECG signals as training data for a kNN classifier, achieving an accuracy of 87.5% in classifying ten heart diseases (left bundle branch block, right bundle branch block, atrial premature beat, paced rhythm, ventricu-

lar flutter, ventricular tachycardia, ventricular fibrillation, ventricular bigeminy, asystole, and high grade ventricular ectopic activity) and the normal state.^[111] Another study conducted a real-time classification of ECG signals for 24 h into two classes, normal and abnormal.^[110] This classification was accomplished using the discrete wavelet transform (DWT) as the feature extraction method and an SVM classifier, marking an accuracy of 98.9%. Zabihi et al. detected atrial fibrillation by using a random forest classifier to classify the ECG types into four classes (normal, atrial fibrillation, other, and noisy).^[113] After denoising ECG signals and removing baseline wander, 491 hand-crafted features were extracted, which encompassed base-level features and meta-level features, and 150 ranked features were selected based on the reduction of entropy. This approach achieved the first-place tie in the PhysioNet/CinC Challenge 2017 with an overall score of 82.6%.

In addition to the aforementioned machine learning algorithms, researchers have explored deep learning-based models for heartbeat classification. One such model is the feed-forward neural network, which is a type of artificial neural network (ANN) without feedback loops.^[115–117] For instance, Güler et al. decomposed the ECG signals into a time-frequency domain by using the DWT for the feature extraction and then classified the heartbeat into four types (normal, congestive heart failure, ventricular tachyarrhythmia, atrial fibrillation) based on multilayer perceptron (MLP) neural network.^[115] Wang et al. successfully classified the eight different types of ECG arrhythmia by using a probabilistic neural network-based classifier.^[116] They conducted feature extraction and normalization on ECG beat samples, followed by feature reduction using principal component analysis with linear discriminant analysis (LDA), resulting in an average classification accuracy of 99.71%.

Convolutional neural networks (CNN) have also gained popularity as an approach for ECG classification.^[118–123] Kiranyaz et al. proposed the use of 1D CNNs instead of conventional 2D CNNs for ECG classification.^[118] They suggested the advantage of 1D CNN is that it can directly process raw data without requiring pre-processing steps such as converting 1D data to 2D, leading to a more compact architecture with minimal computational complexity. On the other hand, Panganiban et al. utilized deep CNNs along with translated 2D spectrogram images derived from converted text files.^[122] By employing this approach, they achieved the successful classification of five types of heartbeats (including four types of arrhythmias and normal) with an average accuracy of 98.73%.

ECG classification using a recurrent neural network (RNN) has emerged as a promising approach for leveraging the temporal dynamics of ECG signals.^[124–127] Yildirim introduced the ECG heartbeat classification method based on Long Short-Term Memory (LSTM), which is one of the most popular types of RNN architecture.^[124] This model incorporates a wavelet-based layer that decomposes the ECG signals into frequency sub-bands, which serve as input sequences for LSTM networks, achieving a high recognition performance of 99.39% in classifying ECG signals into five heartbeat types. Furthermore, there have been several approaches to combining CNN with LSTM for accurate classification.^[128–130] Petmezas et al. proposed a hybrid deep learning model architecture that combines a CNN for feature extraction with an LSTM model for temporal dynamics memorization, enabling accurate classification of four ECG rhythm types,

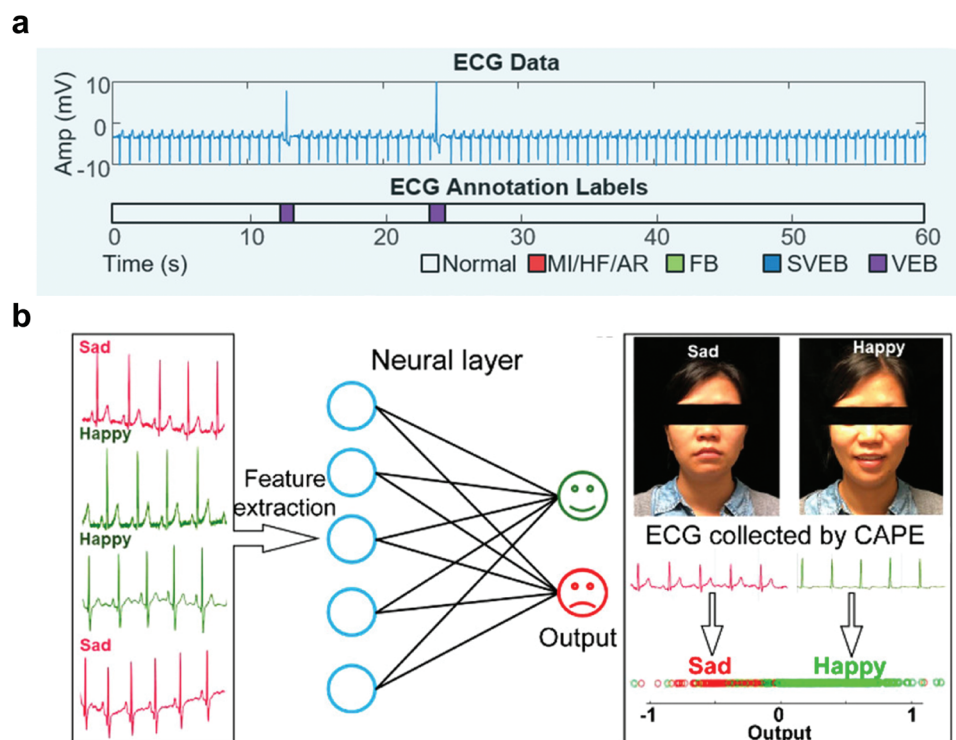


Figure 4. Applications of wearable ECG sensor devices assisted by machine learning techniques. a) Real-time monitoring of ECG data and labeling it into five categories (normal, myocardial infarction/heart failure/miscellaneous arrhythmia, fusion beat, supraventricular ectopic beats, and ventricular ectopic beats) based on sequence-to-sequence annotation approach. Reproduced under the terms of the CC-BY license.^[78] Copyright 2019, the authors, published by John Wiley and Sons. b) Classifying the emotional states (happiness and sadness) using ECG signals with an RNN methodology. Reproduced with permission.^[83] Copyright 2020, American Chemical Society.

including atrial fibrillation.^[129] By utilizing focal loss to handle data imbalance, the model achieved high sensitivity (97.87%) and specificity (99.29%) in detecting common atrial fibrillation in ECG signals.

In addition to these approaches, several works were using deep-learning-based algorithms for the ECG heartbeat signal classification. Xia et al. proposed an automatic ECG classification system based on the stacked denoising autoencoder with sparsity constraint and softmax regression.^[60] They successfully classified the ECG data from both the MIT-BIH database and the wearable sensor device. Kim et al. developed the deep learning network architecture designed for the annotation of ECG data, enabling precise labeling of heartbeats into five categories (normal, myocardial infarction/heart failure/miscellaneous arrhythmia, fusion beat, supraventricular ectopic beats, and ventricular ectopic beats) (Figure 4a).^[78] By adopting a sequence-to-sequence annotation approach and utilizing existing data, their model showed a difference from previous CNN-based classification methods, delivering precise beat labeling of 98.7% average accuracy.

The utilization of machine learning and deep learning techniques in ECG analysis allows for more accurate and automated classification of heartbeats, providing valuable insights for clinical diagnosis and treatment decisions. These advancements enable healthcare professionals to identify and address cardiac abnormalities more efficiently, leading to improved patient outcomes and personalized care.

In recent years, machine learning techniques have expanded the scope of ECG analysis beyond heartbeat classifications, enabling the detection and prediction of various diseases. For instance, stroke prediction has been explored by leveraging machine learning models such as random trees,^[131] CNN,^[132] and LSTM.^[133] These models analyze ECG signals to identify patterns associated with stroke risk, providing valuable insights for early intervention and prevention. Additionally, machine learning algorithms have been employed to predict patient-specific seizure occurrences based on ECG signals. Using SVM, researchers successfully classified preictal and interictal phases, achieving an average prediction time of 13.7 minutes with a sensitivity of 89.06%.^[134] Moreover, Wang et al. introduced a modified LeNet-5 CNN approach for detecting sleep apnea by analyzing ECG signals.^[135] In addition to labeled segments, they also incorporated adjacent segments, resulting in an improved average accuracy of 97.1%. By analyzing ECG signals, these machine learning-based approaches contribute to the early detection and prediction of diseases, enhancing patient care and treatment outcomes.

Another intriguing application in the realm of health care and biomedical engineering is ECG-based stress detection.^[136–139] Keshan et al. utilized machine learning algorithms, including a decision tree classifier and Naïve Bayes, to analyze ECG signals for stress detection in automobile drivers. Their approach achieved an accuracy of 88.24% in classifying three stress levels (low, medium, and high) and 100% accuracy in distinguishing

between low and high-stress levels.^[136] In a different study conducted by Smets et al., five-day ambulatory monitoring of stress was performed using wearable devices that captured physiological information, including skin conductance, skin temperature, acceleration, and ECG signals.^[137] Through a large-scale cross-sectional study involving 1002 subjects, they identified digital phenotypes associated with self-reported poor health indicators and high depression, anxiety, and stress scores. By employing a random forest model, they successfully correlated these physiological signals with daily-life stress levels, highlighting the potential of ECG and other physiological signals for stress detection in real-life settings.

Moreover, the analysis of physiological signals, such as ECG not only provides insights into individuals' stress levels but also explores the broader domain of human emotion recognition.^[83,140–142] RNN was combined with continuous recordings of ECG to distinguish emotional states (Figure 4b).^[83] The system achieved competitive results (f-score = 0.73) in classifying positive and negative feelings, demonstrating its potential application in smart health monitoring and management fields. Sepúlveda et al. presented an improved approach for emotion recognition using ECG signals by employing wavelet transform for signal analysis.^[141] Features extracted from the ECG signal using a wavelet scattering algorithm at different time scales were fed into various classifiers, such as decision tree, LDA, Naïve Bayes, kNN, SVM, and ensemble. The results demonstrate superior performance with an accuracy of 88.8% in valence, 90.2% in arousal, and 95.3% in two-dimensional classification, surpassing previous studies.

Beyond its traditional diagnostic role, ECG analysis has shown promise as a biometric identifier, where machine learning algorithms can be employed to extract unique features from ECG signals, enabling accurate human identification for security and authentication purposes. Zhang et al. introduced a novel wearable ECG-based user identification system.^[57] The ECG heartbeats were projected to a 2D state space and transformed into 2D images, while a CNN model automatically learned intricate patterns from these images to perform user identification. Experimental results using a wearable prototype demonstrated promising identification rates of 98.4% (single-arm-ECG) and 91.1% (ear-ECG) on two acquired datasets. Additionally, Lynn et al. proposed a deep RNN based on a Gated recurrent Unit (GRU) for human identification from ECG data.^[143] The bidirectional RNN with GRU cell learned representations from both previous and future time steps, enhancing context understanding and reducing ambiguity. The model achieved an impressive classification accuracy of 98.55%.

4.2. EMG with Machine Learning Techniques

Recent advances in machine learning techniques have opened new avenues for human-machine interfaces (HMI) using EMG sensors. Machine learning algorithms can be trained to classify and predict human motions based on EMG signals. This approach has great potential to be applied in various fields such as rehabilitation, prosthetics, and interfaces for virtual reality.^[144,145]

The integration of wearable EMG and machine learning technology enables the seamless interaction between humans and

machines. Wearable EMG sensors provide continuous signal information, allowing for real-time monitoring of muscle activity. Machine learning techniques applied to the data from these sensors enable the prediction and classification of user movements, facilitating real-time HMI based on the information contained in the EMG signal. Depending on the targeted muscle group of wearable EMG sensor devices, diverse HMI applications have been introduced including hand gesture recognition,^[63,65,91,92,146] artificial throat,^[80,93,94] and lower limb activity analysis.^[147–149]

Hand gesture recognition is a widely employed and significant application of EMG technology in the field of HMI. By utilizing wearable EMG sensors placed on the upper limb muscles, hand gestures can be accurately classified through signal analysis using machine learning techniques. This technology finds diverse applications in fields such as robotics, virtual reality interfaces, and prosthetics. Various classification techniques have been utilized, ranging from classical machine learning algorithms like SVM and LDA to deep learning models such as CNN and RNN. Yang et al. proposed wearable armband EMG devices with an IoT system that classifies nine hand gestures by LDA, MLP, and SVM.^[63] They classified the hand gestures in 96.20% in real-time to operate the five-finger robotic hands mimicking the users' hand motion. Another study by Lee et al. utilized the stretchable array EMG sensor to classify 18 hand gestures including both static and dynamic gestures.^[92] They preprocessed the raw EMG signals from eight channels with continuous wavelet transform and then employed the following graph neural network for gesture recognition, achieving 94.82% average accuracy over 72 h. On the other hand, hyperdimensional computing was also applied for the real-time classification of 21 hand gestures using screen-printed uniform 4×16 arrays of electrodes on a flexible PET substrate (Figure 5a).^[146] This approach highlighted the efficiency and adaptability of this method across various situational contexts such as different arm positions. In addition to hand gesture recognition, a multitude of machine learning-based methodologies have been developed for the recognition of upper limb movements. These methodologies encompass a wide range of applications, including the tracking of 3D finger joint angles,^[64] classification of arm movement patterns,^[67] identification of wrist movements,^[150] and handwritten character recognition.^[151]

The development of an artificial throat based on facial EMG signal represents a novel and pioneering application within the domain of HMI. This approach offers a potential solution to patients with limited vocalization abilities. The muscles near the larynx and low jaw produce EMG signals that carry pertinent voice-related information essential for speech.^[152] By employing advanced machine learning techniques, these EMG signals can be effectively analyzed and interpreted, enabling accurate prediction and classification of the intended words or phrases desired by the users. Liu et al. present a study on a wearable EMG-based speech recognition system as an HMI by successfully classifying the command signals (Left, Right, Up, Down) for Pacman games.^[80] By employing LDA as a classifier, they achieved a 90% recognition accuracy for a real-time HMI application. Another research utilized LDA as a classifier for action instruction and emotion instruction, recording three EMG signals from different muscle groups (lower jaw, left and right sides of the face) using tattoo-like EMG patches. They employed wavelet

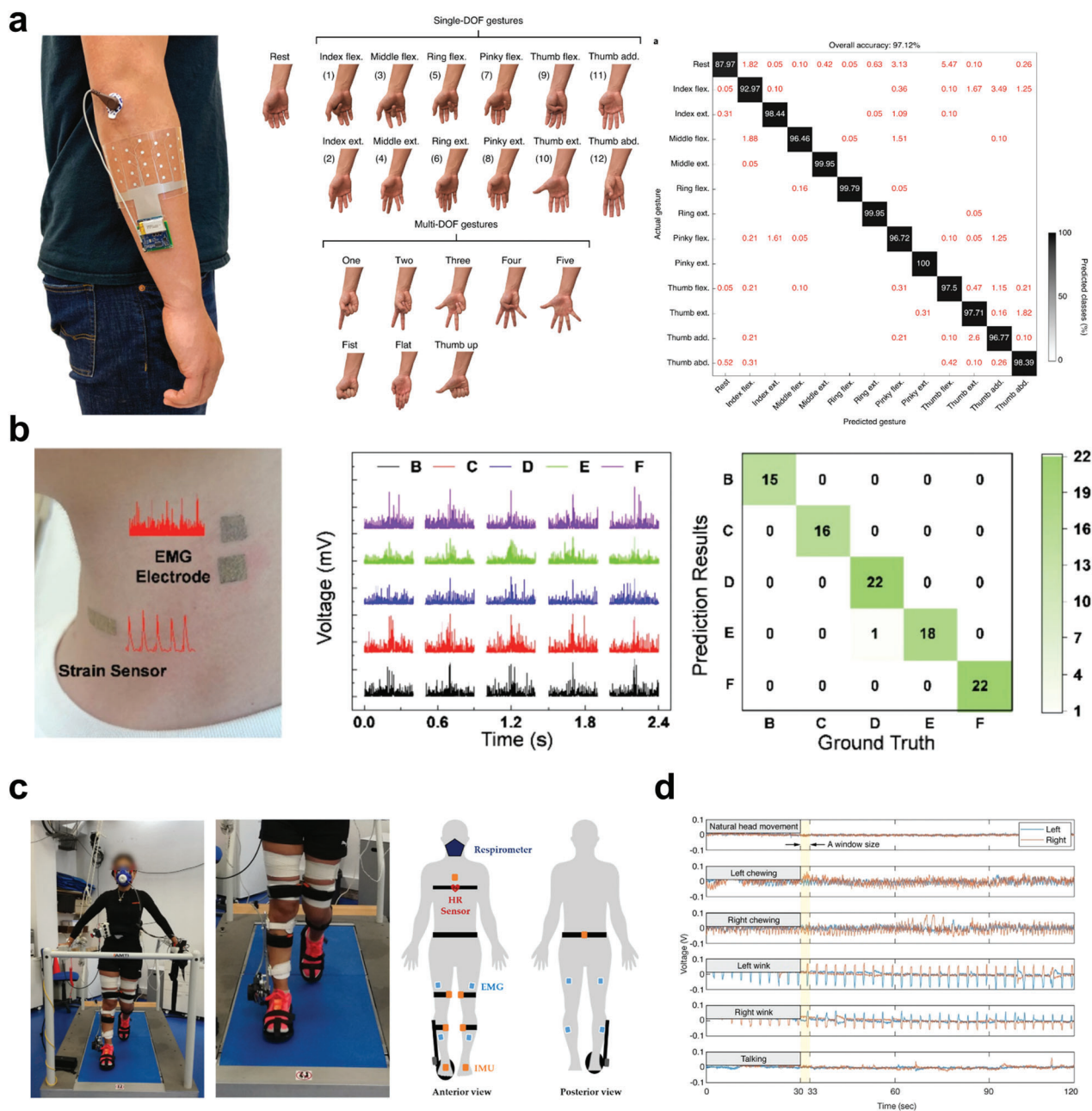


Figure 5. Wearable EMG sensor devices and their applications incorporated with machine learning techniques. a) Wearable EMG electrode arrays for hand gesture recognition for the real-time classification of 21 hand gestures. The confusion matrix showed the 97.12% accuracy of real-time classification of single-DOF hand gestures during 4 s hold periods of baseline context testing. Reproduced with permission.^[146] Copyright 2020, Springer Nature. b) E-skin incorporating EMG electrode with a strain sensor for the artificial throat, distinguishing five letters (B, C, D, E, F). Detected EMG signals corresponding to different letters. The confusion matrix shows a 98.9% classification accuracy in distinguishing five letters, achieved through the CNN approach using signals from EMG electrodes and strain sensors. Reproduced with permission.^[94] Copyright 2022, Elsevier. c) Lower limb EMG sensor setup during walking trial for estimating the energy expenditure in assisted or non-assisted gait. Reproduced under the terms of the CC-BY license.^[154] Copyright 2022, the authors, published by MDPI. d) EMG signals for distinct face activity patterns such as chewing, winking, and talking, based on a glass-type EMG sensor. Reproduced under the terms of the CC-BY license.^[70] Copyright 2017, the authors, published by Springer Nature.

transformation on EMG signals to reduce noise, subsequently extracting crucial features including wavelet coefficient pairs, fourth-order autoregressive model coefficients, and the initial two cepstral coefficients to effectively characterize each trial. The study reported average accuracies of 92.33% and 89.04% for each

instruction, respectively.^[93] Incorporating an EMG electrode with a strain sensor, Qiao et al. distinguish the five letters (B, C, D, E, F) with an accuracy of 98.9% (Figure 5b).^[94] They fabricated an Au/Pu nanomesh as a strain sensor for detecting throat vibration and an Au/PVA nanomesh as an EMG sensor. A synergistic

CNN algorithm was employed to process the combined EMG-strain signals, consisting of a 1D ResNet18 for the EMG part and a two-layer CNN for the strain part.

The applications of lower limb EMG signals combined with machine learning techniques have been showing tremendous potential in various areas. In the area of HMI, lower limb EMG signals have been analyzed for the recognition of foot gestures,^[147] or gait event prediction.^[148–149] Maragliulo et al. successfully utilized wearable foot band EMG sensors with two channels, capable of classifying five-foot gestures using an SVM classifier.^[147] They did the 18-session experiments with three subjects, showing the robustness and accuracy of the device. This approach highlights the potential of the device's usage in the hand-free controller such as foot pedals. Gait phase classification in natural ground walking conditions was conducted by Morbidoni et al. by analyzing eight lower limbs EMG with an ANN-based approach.^[148] They classified the 23 subjects' gait events in stance or swing phases and predicted the transition between phases such as heel strike and toe-off. Furthermore, lower limb EMG has yielded significant insights in the domains of rehabilitation and clinical trials, in addition to its application in HMI. For instance, the crouch gait pattern caused by foot drop disorder of patients was predicted based on leg EMG signals, analyzed by an extreme learning machine.^[153] Another research focused on estimating the energy expenditure in assisted or non-assisted gait, which is a crucial marker in gait rehabilitation (Figure 5c).^[154] This estimation was accomplished based on eight-channel leg EMG with other inertial data, anthropometric data, and heart rate, employing deep learning regressors such as LSTM and CNN. These studies highlight the considerable potential of integrating lower limb EMG sensors with machine learning techniques to enhance various aspects of HMI, rehabilitation, and clinical assessments.

Wearable EMG sensors have been offering a wide range of applications beyond HMI applications, demonstrating their versatility when integrated with machine learning techniques. Notably, they have shown promise in the enhanced diagnosis of conditions such as blepharospasm^[155] and bruxism,^[156] where accurate classification of clinical symptoms is achieved by analyzing EMG signals from the orbicularis oculi and jaw muscles. Additionally, EMG-based methods have been employed to differentiate Parkinson's disease from essential tremor, enabling early detection and providing a valuable reference for diagnosis.^[157]

Facial EMG analysis combined with machine learning techniques has been offering diverse applications. One significant role of this is emotion recognition, which gives valuable insights into an individual's emotional state by capturing and interpreting facial muscle activity.^[158–161] For instance, Perusquía-Hernández et al. successfully detected subtle smiles in response to advertisement videos.^[160] They classified micro-smile, which lasts within a second, from other expressions (no-expression, smile, and laughter) by utilizing a neural network with one hidden layer of four Sigmoid neurons. These approaches hold great potential for psychological and neurological research and objective emotion assessment in clinical surveys. Additionally, Facial EMG was utilized for classifying face activity patterns such as chewing, winking, and talking, by employing SVM classification with a glass-type EMG sensor (Figure 5d).^[70] Moreover, Huang et al. extracted four features from EMG signals of every single chew and utilized a decision tree-based classifier for classifying five

food types.^[71] This approach holds promise for long-term dietary analysis and nutritional monitoring by categorizing food types.

These diverse applications highlight the expanding potential of wearable EMG sensors combined with machine learning techniques in various fields such as improved diagnostics, emotion recognition, and dietary analysis. This integration of EMG and machine learning holds significant promise for pioneering into other healthcare applications and enhancing individuals' quality of life.

4.3. EEG with Machine Learning Techniques

EEG is a non-invasive technique used to measure and record the brain's electrical activity. Traditionally, EEG has been employed in clinical settings for diagnosing and monitoring neurological disorders, such as epilepsy and sleep disorders. However, recent advancements in machine learning have unlocked novel possibilities for analyzing and interpreting EEG data beyond its traditional clinical applications. Machine learning algorithms now enable the analysis of brain signals to detect patterns associated with specific tasks, cognitive functions, and emotional states. This has led to diverse applications in fields such as psychology, neurology, human-computer interaction, marketing, and advertising. Moreover, EEG-powered machine learning models hold great promise in the realm of brain-computer interfaces (BCIs), allowing individuals to control external devices or communicate directly with computers using brain activity. BCIs offer significant benefits to people with severe physical disabilities, granting them independence and improving their quality of life.

The success of EEG powered by machine learning depends on the availability of extensive and well-curated datasets and the wearability of EEG devices. These datasets serve as training material for machine learning algorithms, enabling them to understand the complex relationships between EEG signals and corresponding mental states or actions. With the increasing availability of data and continuous advancements in machine learning algorithms, the potential for EEG-powered applications is poised to grow exponentially. In conclusion, EEG powered by machine learning represents a transformative approach to understanding and harnessing the capabilities of brain activity. By combining the richness of EEG data with the computational prowess of machine learning algorithms, we can unlock novel insights into the human mind and develop innovative applications that have the potential to revolutionize healthcare, communication, and our overall comprehension of the brain.

Advancements in technologies, such as epidermal materials, processing power, and wireless technology, have significantly improved the wearability of EEG devices. This revolution in EEG technology has provided non-intrusive, mobile, and user-friendly solutions for brain activity monitoring. Nevertheless, EEG data obtained from external skin surface recording is prone to various artifacts, which can hinder accurate characterization and classification. Signal processing plays a crucial role in ensuring clear and reliable EEG interpretation. Machine learning serves as a potent tool for effective noise reduction and has been employed in EEG studies.^[162,163] For instance, therapeutic neuro-feedback using EEG is essential, but the presence of artifacts can limit its effectiveness.^[162] To address this, a study introduced a

self-calibrating protocol that combines five standard machine learning algorithms to classify brain states associated with “pain” or “no pain.” SVM, complex decision trees, kNN, and 2-layer perceptron neural networks were utilized as machine learning classification algorithms. The protocol, implemented on wearable EEG sensors, yielded sufficient data accuracy to reliably distinguish between these two contrasting brain states. Similarly, the field of epilepsy monitoring faces challenges in identifying EEG artifacts that may be mistaken for actual seizures due to similar appearances in amplitude and frequency.^[163] To tackle this issue, a study proposed an artifact detection algorithm designed for a parallel ultra-low-power embedded platform. They used the Tree-based Pipeline Optimization Tool as an automated machine learning system, resulting in long-term epilepsy monitoring solutions with low false-positive rates and high sensitivity.

Powered by machine learning and wearability, EEG devices could be used for the detection or prediction of variable diseases.^[164,165] For instance, seizure, which is a phase characterized by symptoms arising from unusually high or synchronized neural activity in the brain, is essential for evaluating and treating epilepsy. Currently, epilepsy care relies on seizure diaries, but this method may overlook some seizures during clinical monitoring. Thus, wearable EEG devices could offer a promising alternative for long-term continuous monitoring. Machine learning algorithms could be utilized for more accurate data collection, classification, and prediction of the disease. Tang et al. employed epilepsy monitoring EEG units with multisignal biosensors. The sensor acquired EEG, PPG, body temperature, and acceleration.^[166] The machine learning algorithms were trained and tested using two distinct approaches. Algorithm 1 involved machine learning techniques to create specific detection models for nine different seizure types. On the other hand, Algorithm 2 utilized machine learning methods to develop a general detection model that grouped all seizure types, irrespective of their specific characteristics. Both algorithms were based on CNN to categorize raw time series data, and to ensure a balanced dataset, random undersampling was applied to the raw data. Algorithm 1 demonstrated successful detection for eight out of the nine seizure types, outperforming chance levels (AUC-ROC: 0.648 to 0.976). Meanwhile, Algorithm 2 showed successful detection for all nine seizure types, surpassing chance levels (AUC-ROC: 0.642 to 0.995). The fusion of the accelerometer and blood volume pulse modalities yielded the highest AUC-ROC of 0.752 when considering all seizure types combined. Similarly, a novel seizure predictive model utilizing Ear EEG, ECG, and PPG signals, collected through a device suitable for static and outpatient use, was developed (Figure 6a).^[167] By employing supervised machine learning techniques and processing this data, various predictive models were developed to classify the epileptic person's state as normal, pre-seizure, or seizure. Through validation, a refined model based on Boosted Trees achieved a high prediction accuracy of 91.5% and sensitivity of 85.4%. Rapid detection and prediction of multiple cognitive impairments is crucial, however, conventional methods like MRI are expensive and involve intricate analysis. As an alternative, researchers have been exploring Alzheimer's disease detection methods utilizing machine learning and EEG in recent years.^[168] A study presents a preliminary assessment of a self-driven multi-class discrimination approach based on a commercially available EEG acquisition system with

sixteen channels. The process involved the evaluation of a multi-class classification problem using a MLP with leave-one-subject-out cross-validation. The preliminary findings showed promising results, comparable to the best existing literature (0.88 F1-score). This suggests that Alzheimer's disease could potentially be detected using a self-driven approach based on a commercial EEG and machine learning. In the same way, another study was conducted to identify the optimal channel configurations for wearable EEG devices to facilitate computer-aided diagnosis of mild cognitive impairment.^[169] SVM was employed for feature selection and classification, and a leave-pair-out cross-validation algorithm was used for the evaluation of classification accuracy. Subsequently, the electrode configurations yielding the highest diagnostic accuracy were recommended for each condition.

Furthermore, thanks to state-of-the-art mathematical algorithms, advanced information can be gathered with EEG, such as emotion, stress, and mental health.^[170] As an example, the focus on brain activity associated with sustained attention has been the subject of numerous neurophysiological studies. Especially, for safe driving, it is crucial to evaluate driver vigilance with an EEG system. The research introduced a new system based on an EEG sensor-based mobile wireless system for measuring driver vigilance.^[171,172] The system allowed real-time monitoring of a driver's vigilance status, enabling a correlation between fluctuations in driving performance and changes in brain activities. The system processed EEG recordings and translated them into vigilance levels with the SVM algorithm. In the same way, it is important to detect workers' conditions for a safe work environment. Especially, Construction is renowned for being one of the most demanding occupations. The significant impact of stress on workers' productivity, safety, well-being, and work quality underscores the importance of managing excessive stress. Thus, a study proposed a machine learning-based system for identifying construction workers' stress, based on their brain activity recorded by a wearable EEG. The results demonstrated an encouraging 71.1% accuracy using SVM in recognizing workers' stress.^[173] Similarly, another approach was conducted by simulating hazards in construction sites on VR (Figure 6b).^[174] The complex machine learning algorithms based on the LightGBM algorithm were utilized for more accurate classification. Aside from construction sites, other researchers developed EEG instruments for detecting workers' stress by frontal asymmetry monitoring.^[175] The single-channel differential measurement focuses on analyzing frontal asymmetry, a well-established EEG feature commonly associated with stress assessment. To classify stress levels, four well-known machine learning algorithms (SVM, kNN, random forest, and ANN) were trained on 50% of the dataset. Remarkably, they achieved more than 90% accuracy in classifying each 2-second epoch of EEG data acquired from the stressed subjects. Furthermore, the capabilities of wearable EEG devices could reach for recognizing the mental states of the subjects. A study conducted experiments using two different types of video input, instructional and recreational, then measured EEG signals by commercially available wearable devices.^[176] The researchers utilized various machine learning methods such as SVM, sparse logistic regression, and Deep Belief Networks to distinguish between the mental states (logical and emotional) induced by the two types of video input. Moreover, more advanced emotion classification is available through VR-EEG.^[177]

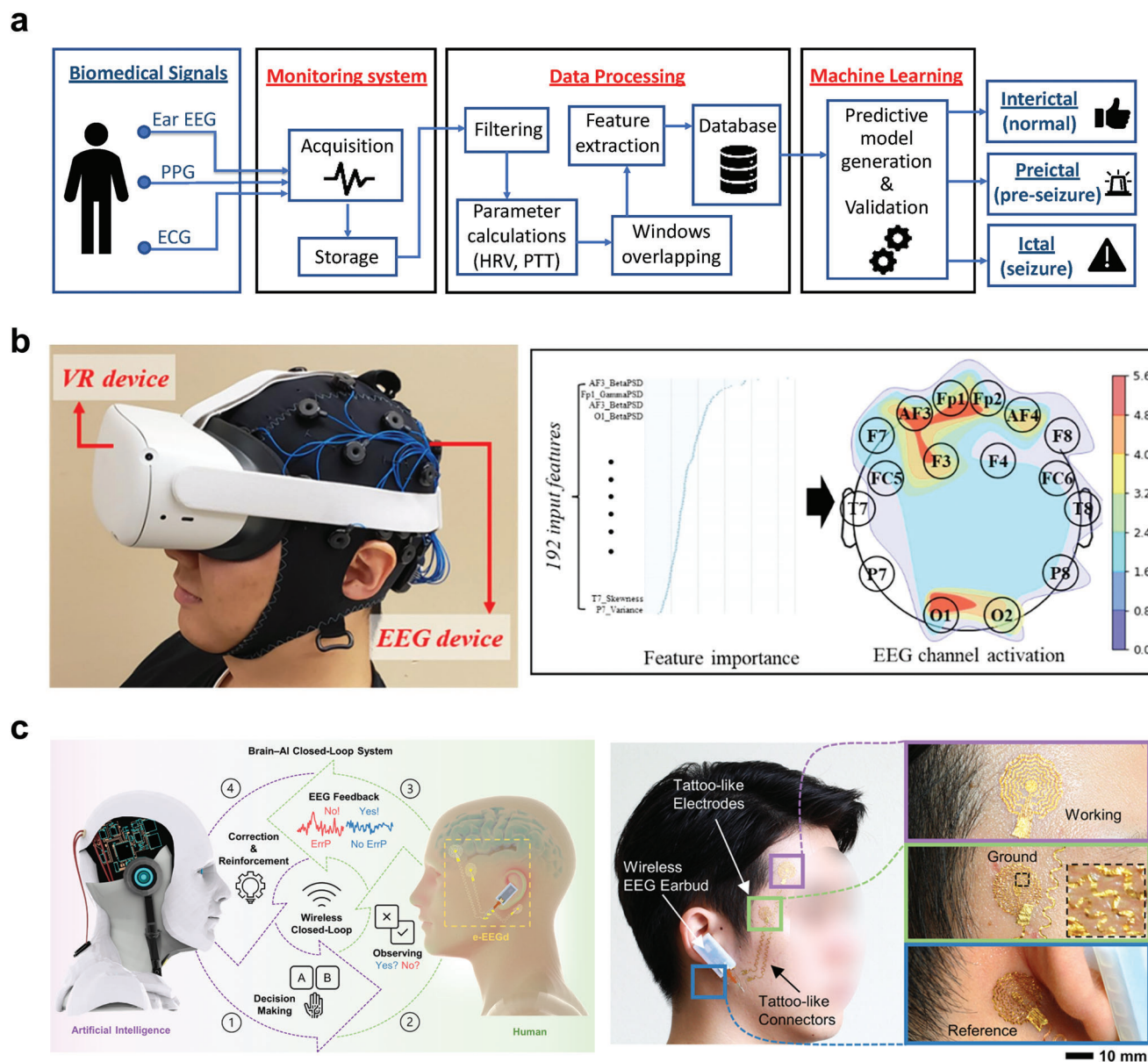


Figure 6. The utilization of wearable EEG sensor devices alongside machine learning methods in various applications. a) A system for predicting epileptic seizures using machine learning techniques, incorporating signals from ECG, PPG, and EEG sensors in wearable devices. Reproduced under the terms of the CC-BY license.^[167] Copyright 2022, the authors, published by MDPI. b) Emotional state classification using wearable VR-EEG headsets. Reproduced with permission.^[174] Copyright 2022, Elsevier. c) The notion of the Brain-AI Closed-Loop System (BACLoS) and visuals of wearable EEG gadgets comprised of tattoo-like electrodes. Reproduced under the terms of the CC-BY license.^[97] Copyright 2022, the authors, published by Springer Nature.

This study focused on inducing four emotions (happy, scared, calm, and bored) using a VR headset and collecting brainwave samples through a low-cost wearable EEG headset. Multiple machine learning algorithms were used, and among them, SVM achieved an impressive 85.01% accuracy.

BCI is prominent due to its direct brain communication, accessibility benefits for individuals with disabilities, medical applications, improved human-computer interaction, and cognitive enhancement potential. Machine learning benefits BCI by enabling accurate pattern recognition in brain signals, adapting the inter-

face to individual users, and efficient feature extraction, leading to improved brain-computer interaction and usability. Thus, Shin et al. developed a Brain-AI Closed-Loop System (BACLoS) based on a wearable EEG system.^[97] In this study, they developed a wireless earbud-like EEG device, along with tattoo-like electrodes and connectors (Figure 6c). This combination allows continuous recording of high-quality EEG signals, with a focus on the error-related potential (ErrP). Tattoo-like electrodes were applied as three electrodes (working, counter, and reference) and connected with tattoo-like connectors to wireless EEG earbuds. The

tattoo-like electrode possesses flexibility akin to human skin, ensuring it can easily conform to rough and uneven skin surfaces. This close conformance significantly reduces the gaps between the skin and the electrodes, resulting in low electrode-skin impedance similar to standard commercial electrodes, all without the need for wet electrolytes. The sensor captures the ErrP signals, which reveal the cognitive impact of an unexpected machine response on the human user. The AI then adjusts its decisions based on the presence or absence of these ErrP signals, determined through deep learning classification of the received EEG data. For deep learning, they adopted deep neural networks and LSTM systems for classification. For demonstrations of BACLoS, they applied it to brain-machine controlling, such as autonomous driving vehicles, maze solvers, and assistant interfaces.

4.4. PPG with Machine Learning Techniques

Machine learning has improved the analysis of PPG signals, resulting in more accurate algorithms for signal processing and feature extraction. As a result, PPG signals can provide a broader range of physiological information, including heart rate, blood oxygen saturation, and blood pressure. Furthermore, machine learning models can detect and classify abnormal symptoms through PPG signals, aiding in the early detection of cardiovascular or respiratory disorders and ultimately improving patient outcomes. Therefore, the use of machine learning algorithms in conjunction with PPG signals has significant potential for improving the diagnosis of cardiovascular and respiratory disorders with further potential for speculating various biosignals.^[167,178,179]

The utilization of PPG initially focused on detecting changes in blood volume in vessels, making it suitable for detecting blood loss.^[180] Accurately assessing the severity of occult hemorrhage is crucial for trauma patients, as severe blood loss (exceeding 30% of blood volume) can result in lethal shock. In this regard, two subject groups were examined: one comprised of volunteers who underwent intentional minor bloodletting, and the other consisted of emergency patients. Custom wearable multi-channel pulse oximeters were employed for data collection. By employing a machine learning algorithm, the two groups were effectively differentiated with an accuracy rate exceeding 80%. During the cardiac cycle, blood volume fluctuates due to pressure variations, which can be detected using PPG and translated into heart rate. Determining the heart rate involves identifying signal peaks from the PPG waveform. However, accurate measurement of heart rate is often impeded by noise and motion artifacts. In a particular study, researchers successfully employed a machine learning approach to differentiate between noise and valid signals.^[43,180,181] Specifically, random forest regression, an algorithm that constructs a virtual forest with numerous decision trees, was employed. Through training with actual data, the algorithm generated refined data while significantly reducing motion artifacts. In addition to random forest regression, another viable approach for artifact detection is supervised machine learning.^[182] Unlike random forest regression, which is an unsupervised algorithm, supervised machine learning relies on labeled data for training. PPG can also be applied in the detection of cardiovascular disorders such as arrhythmia, with the

aid of machine learning algorithms.^[183–185] Arrhythmia refers to irregularities in the heart's rhythm, characterized by abnormal electrical activity. Early identification of these irregularities is critical as they can potentially lead to lethal shock. Artificial intelligence not only facilitates the removal of signal artifacts but also enables the identification of arrhythmias.^[185] By utilizing an Elastic Net model in commercially available wristband smartwatches, researchers achieved a 95% accuracy rate in detecting atrial fibrillation, a common form of arrhythmia. Moreover, post-cardiac surgery patients require careful monitoring of their condition.^[183] By leveraging PPG data obtained from commercial smartwatches, a machine learning algorithm achieved an accuracy rate exceeding 94% in detecting atrial fibrillation, comparable to the performance of multi-lead Holter devices.

Machine learning technology bears even more possibility for PPG, such as blood pressure estimation and cardiovascular disease detection. As a well-known approach, blood pressure could be estimated through the calculation of pulse transit time by comparing ECG and PPG signals.^[186–188] For instance, the wearable ear-ECG/PPG sensor detected heart rate while estimating blood pressure by utilizing machine learning.^[186] The authors proposed situating sensors behind both ears to enhance wearability to successfully capture weak ear-ECG/PPG signals. They utilized machine learning for the removal of motion artifacts and blood pressure estimation. As a first stage, for signal processing, supervised learning by an SVM classifier was applied. Then, unsupervised learning was undergone for signal quality labeling and purification. Heart rate was easily measured through purified signals, while blood pressure was estimated through the measurement of pulse transit time and supervised regression model learning. Meanwhile, other researchers attempted to estimate blood pressure with a different approach. Without both using PPG and ECG, pulse transit time could be deducted by two wearable PPG sensors in different locations of the body.^[189] They employed SVM and linear regression analysis for three layers of the ANN network. Similarly, with PPG signal data, prehypertension, stage 1 hypertension, and stage 2 hypertension could be classified with machine learning algorithms.^[188] Other cardiovascular health signals could be detected through machine learning-powered PPG. A notable heart disease, hypertrophic cardiomyopathy (HCM), which is a disease in which the heart muscle becomes thickened, was able to be discovered with a wearable biosensor (**Figure 7a**).^[190] The disorder often goes undiagnosed because many patients with the disease have few symptoms. In this research, PPG-ECG integrated wearable devices were utilized for the study, and they obtained PPG and ECG signals from 19 HCM patients and 64 healthy volunteers as a control cohort. The multiple-instance learning via embedded instance selection method was applied as an automated classifier algorithm, with an evaluation of Leave-One-Group-Out cross-validation. The analysis demonstrated significant anomalies in pulse waves, such as systolic ejection time, rate of rise during systole, and respiratory variation. Thus, the study achieved the effective classification of HCM at 99% for the patient, showing the potential for a noninvasive, wearable home-care diagnostic device. Arterial stiffness, which is a parameter strongly associated with cardiovascular wellness, could also be identified with one ECG and one PPG wearable module.^[191] Simultaneously collecting ECG and PPG signals, the sensor extracts 21

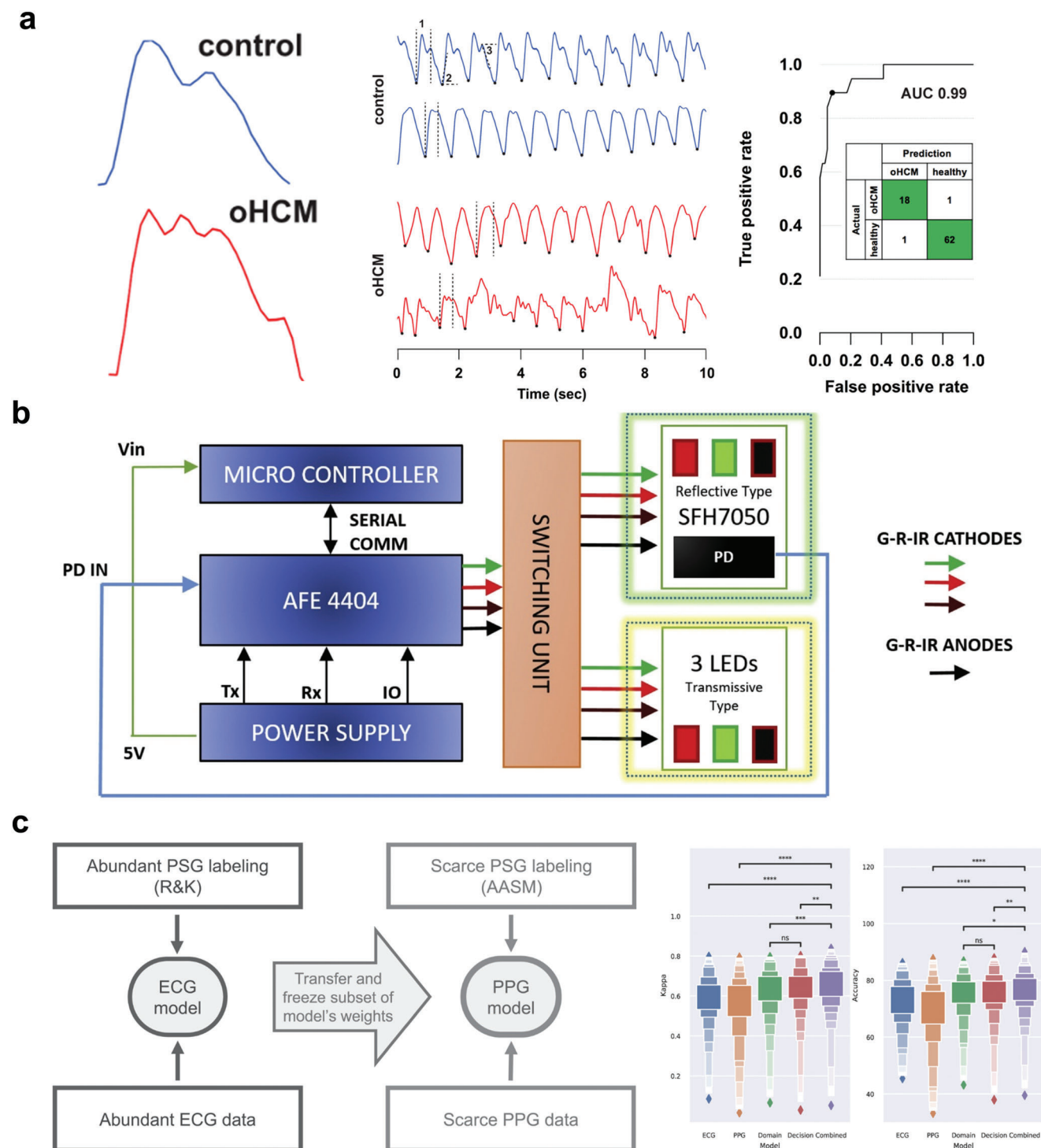


Figure 7. The application of wearable PPG sensor devices in combination with machine learning methods across multiple applications. a) Differences observed in PPG tracings between patients with obstructive hypertrophic cardiomyopathy (oHCM) and healthy volunteers. Continuous PPG signals recorded over 10 s from two healthy volunteers and two oHCM patients are depicted. The confusion matrix and receiver-operator curve are presented. Reproduced under the terms of the CC-BY license.^[190] Copyright 2019, the authors, published by Springer Nature. b) Block diagram of PPG-based wearable blood glucose detection sensor. Three different LEDs were utilized in both reflective and transmissive types. Reproduced with permission.^[195] Copyright 2021, Elsevier. c) Schematic of transfer training algorithm with superior Kappa value and accuracy of the combined algorithm. Reproduced under the terms of the CC-BY license.^[200] Copyright 2021, the authors, published by Springer Nature.

features to evaluate arterial stiffness. A feature selection method based on genetic algorithms is used to identify crucial indicators. The model is developed using multivariate linear regression (MLR), decision tree, and backpropagation (BP) neural network techniques. The BP neural network-based approach outperforms results from conventional arteriosclerosis instruments in estimating cardiovascular disease risk (correlation coefficient: 0.9488; mean residual: -0.3579% ; standard deviation of residual: 3.7131%). These findings highlight the considerable potential of the proposed machine learning-based wearable PPG sensor in monitoring cardiovascular wellness with increased wearability and cost-effectiveness.

Machine learning has opened up even more possibilities of advanced information for PPG such as blood glucose, mental status, sleep, etc.^[192,193] Measuring blood glucose levels is crucial for diabetic individuals to regularly monitor their levels to avoid severe acute complications. Recent advancements in noninvasive optical signals, such as PPG, have demonstrated their effectiveness in measuring human physiological and vascular conditions and estimating blood glucose levels. To explore this further, a study with a clinical trial was conducted involving nine type-2 diabetic patients who used wearable devices to acquire PPG signals.^[194] Subsequently, global and personalized models were constructed using random forest regression and adaptive boosting machine learning modeling. Another study was conducted with the pulse oximeter to measure blood glucose levels alongside other physiological indicators like heart rate and blood oxygen percentage.^[195] In the study, the fingertip PPG device was developed, which integrates both transmissive and reflective data acquisition systems, utilizing red, green, and IR LEDs to illuminate the skin (Figure 7b). From the PPG signals obtained, specific and relevant features are extracted, and machine learning algorithms (random forest regression and XGBoost) are applied to predict the actual blood glucose values based on these features. The proposed algorithm and system exhibit a high level of clinical accuracy in predicting blood glucose levels. Detecting stress-related issues early is crucial to prevent mental and physical health, and continuous monitoring of stress is important.^[196] To achieve stress monitoring, a paper proposed a layered system architecture that allows the collection of data samples for labeling.^[197] Binary stress detection based on heart rate and heart rate variation using machine learning methods was investigated. The binary stress detector demonstrates reasonable accuracy in distinguishing between stressful and non-stressful samples, achieving a macro-F1 score of up to 76%. Similarly, three physiological signals, namely electrodermal activity, ECG, and PPG, all of which can be gathered through smartwatches, could be utilized for stress classification.^[198] Also, PPG could contribute to unobstructed home sleep monitoring and management.^[199] However, PPG data is seldom acquired in large sleep studies that rely on polysomnography, which mainly collects ECG data. Radha et al. presented abundant-ECG-data-based transfer training of scarce-PPG-data in a deep RNN (Figure 7c).^[200] The neural network was initially trained on a large sleep dataset containing ECG data from 292 participants and 584 recordings to perform 4-class sleep stage classification (wake, rapid-eye-movement, N1/N2, and N3). The researchers then perform transfer learning to adapt a portion of the neural network's weights to a smaller, more recent PPG dataset comprising 60 healthy par-

ticipants and 101 recordings. The transfer training method outperformed the PPG-trained and ECG-trained baseline models. The performance achieved for PPG-based 4-class sleep stage classification is unprecedented in existing literature, moving home sleep stage monitoring closer to clinical use.

5. Discussion

Wearable biosensors, such as ECG, EMG, EEG, and PPG sensors, have emerged as promising tools for monitoring various physiological signals and health conditions in real-time. These sensors can be integrated with IoT systems and machine learning algorithms to enable remote and personalized healthcare and clinical applications. However, to realize the full potential of this integration, a comprehensive strategy is imperative.

First and foremost, the comfort and wearability of these wearable devices are pivotal aspects. Users must feel at ease while wearing them, especially when continuous monitoring over extended periods is essential. The comfort level directly affects user compliance and the accuracy of data collected. Moreover, comfort is inherently connected to the portability of these devices. Wearers should be able to effortlessly integrate them into their daily routines. Achieving this level of portability hinges on minimizing the size and weight of the devices. Smaller, lightweight wearables tend to be more comfortable and practical for users, as they don't impede mobility or become burdensome during use. However, the challenge arises when attempting to reduce the size and weight of these devices without compromising their functionality, particularly regarding the size of the battery. Batteries power these wearables, and there's often a trade-off between device form factor and battery capacity. Smaller batteries may lead to reduced operational time, potentially inconveniencing users who expect long-lasting, continuous monitoring. To address this trade-off, efficient power management techniques play a critical role. Power-efficient design and management can optimize how the device uses its battery power. For instance, sensors can be programmed to gather data selectively or in response to specific events, thereby minimizing unnecessary power consumption.^[201] Optimizing data transmission, including factors like transmission frequency, data compression, and wireless communication standards, is also crucial for efficient power management in wearable devices.^[202] Otherwise, a strategy such as adding an energy harvesting part or wireless charging design could be adopted. Incorporating an energy harvesting part could extend battery life by utilizing ambient energy sources, reducing the need for large batteries, and improving comfort and portability.^[203] However, its effectiveness depends on available energy sources and may add design complexity. Meanwhile, wireless charging provides convenience and user-friendliness but relies on dedicated infrastructure and may be less energy efficient.^[99,100] The choice between these technologies should align with specific use cases and design goals.

Another important part is to ensure the long-term reliability and durability of wearable devices, particularly skin-adhered patch sensors. Challenges such as maintaining adhesion in diverse environmental conditions and mitigating increased signal noise over time necessitate material and conceptual innovations. Also, considering biocompatibility and breathability in device

materials is essential to ensure skin-friendliness and long-term wearability of wearable devices.

As biosensors deal with sensitive health-related data collected by wearable devices, ensuring robust data security measures is crucial within the IoT framework. Robust encryption protocols, secure data transmission procedures, and strict compliance with data protection regulations are essential elements to safeguard patient information.^[204,205] The sensitivity of health data necessitates a high level of privacy, confidentiality, and protection from unauthorized access. These security measures are fundamental to maintaining trust and ensuring the ethical and legal handling of personal health information in the context of wearable biosensors and healthcare IoT.

6. Conclusion

The integration of machine learning and wearable techniques with four notable biosignals, ECG, EMG, EEG, and PPG, has brought about transformative advancements in various fields of medical monitoring and HMI.

ECG has emerged as a powerful technology for measuring cardiac voltage signals, enabling the non-invasive diagnosis of heart conditions such as arrhythmia. The integration of machine learning techniques has revolutionized ECG analysis, enabling automated heartbeat classification and accurate disease detection. Wearable ECG devices, combined with IoT systems, offer continuous real-time monitoring, while deep learning models enhance classification accuracy. Beyond heartbeat classification, machine learning-based ECG analysis has enabled stroke prediction, seizure occurrence detection, sleep apnea identification, stress detection, emotion recognition, and biometric identification for security purposes. These advancements have significantly improved cardiac healthcare, facilitating early detection, precise diagnosis, and personalized interventions for better patient outcomes.

EMG technology has found wide-ranging applications in sports medicine, rehabilitation, prosthetics, ergonomics, and the diagnosis of neuromuscular disorders. Recent advances in machine learning have opened new possibilities for HMI using wearable EMG sensors. The design of wearable EMG sensors has evolved to enhance wearability and portability, enabling continuous monitoring and analysis of muscle activity. Machine learning algorithms applied to EMG data have facilitated real-time prediction and classification of human motions, leading to practical applications in rehabilitation, prosthetics, and virtual reality interfaces. Hand gesture recognition, artificial throat speech recognition, and lower limb activity analysis are among the significant HMI applications of EMG combined with machine learning. Beyond HMI, EMG with machine learning has shown promise in diagnosing various medical conditions, such as blepharospasm and bruxism, and in differentiating between Parkinson's disease and essential tremor. Facial EMG analysis has also been utilized for emotion recognition, facial activity pattern classification, and dietary analysis. These diverse applications demonstrate the expanding potential of wearable EMG sensors and machine learning in various healthcare domains, promising improved diagnostics, objective emotion assessment, and personalized healthcare solutions for individuals.

EEG powered by machine learning represents a transformative approach to understanding and utilizing brain activity beyond its traditional clinical applications. The integration of machine learning algorithms allows for the analysis of EEG data to detect patterns associated with specific tasks, cognitive functions, and emotional states, opening diverse applications in fields like psychology, neurology, human-computer interaction, marketing, and advertising. Moreover, EEG-powered machine learning models hold great promise in BCIs, offering benefits to individuals with severe physical disabilities and enhancing their quality of life. The success of EEG powered by machine learning depends on the availability of extensive and well-curated datasets and the wearability of EEG devices, which can be further improved with advancements in technologies such as epidermal materials, processing power, and wireless technology.

PPG combined with machine learning has shown remarkable progress in the diagnosis and monitoring of cardiovascular and respiratory disorders. PPG's non-invasive nature, coupled with its convenience and increased accuracy, has made it a valuable tool in medical settings. Machine learning algorithms have enhanced the analysis of PPG signals, leading to more accurate signal processing and feature extraction, thereby enabling a broader range of physiological information to be obtained from PPG signals. Machine learning models have demonstrated the ability to detect and classify abnormal symptoms through PPG signals, offering the potential for early detection of health issues and improved patient outcomes. PPG, powered by machine learning, has been successfully applied in detecting blood loss, measuring heart rate, and identifying arrhythmias, showcasing its versatility in various healthcare scenarios. Moreover, PPG technology holds promise for estimating blood pressure and detecting cardiovascular diseases, further expanding its potential applications in healthcare. The integration of PPG with machine learning has also shown potential in measuring blood glucose levels, monitoring stress, and conducting unobtrusive home sleep monitoring. These cutting-edge advancements in PPG technology, fueled by machine learning, offer exciting prospects for personalized and non-invasive healthcare solutions, ultimately contributing to improved patient care and well-being. As these technologies continue to evolve and mature, their potential for enhancing healthcare outcomes and improving the quality of life for individuals is increasingly evident.

These advancements in machine learning-powered medical technologies offer improved healthcare, early disease detection, and personalized treatments. Moreover, biosignals' potential extends beyond diagnostics, providing insights into detecting emotional states or stress levels and serving as biometric identifiers for security purposes. Further research and technology advancements are crucial to fully realize the potential of these integrated technologies and pave the way for a new era of medical monitoring and HMI.

Acknowledgements

D.K. and J.M. contributed equally to this work. This study was supported by the National Research Foundation of Korea (2021R1A2B5B03001691).

Conflict of Interest

The authors declare no conflict of interest.

Keywords

electrocardiogram, electroencephalogram, electromyogram, machine learning, photoplethysmogram, wearable sensors

Received: July 29, 2023

Revised: September 25, 2023

Published online: October 24, 2023

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